LibFlow
Analysis and Design Document : Assignment 1

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1. Requirements Analysis

1.1 Assignment Specification

The objective of this project is to develop a library management application designed to catalog books and help with user management. The system aims to provide a centralized platform where administrators can manage users (people) and their roles, organize a collection of books with detailed metadata (ISBN, titles), and categorize items through a flexible many-to-many genre classification system.

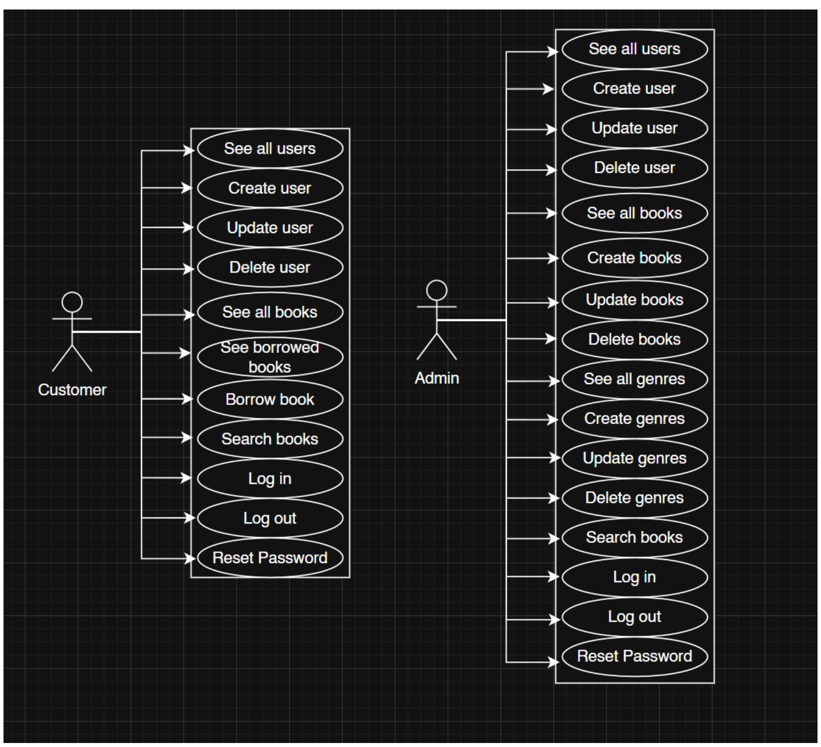
1.2 Functional Requirements

1. Allow the creation, updating, and deletion of user accounts, books and genres.
2. Assign and manage distinct access roles for users (e.g., Customer, Admin).
3. Provide a frontend user interface (UI) for users to interact with the system.

1.3 Non-functional Requirements

1. Implement validation to ensure that all inputs are valid and adhere to the specified format.
2. Use an ORM to handle database interactions.
3. Use a layered architecture for better organization and separation of concerns.
4. Use a database to store all the data.

2. Use-Case Model



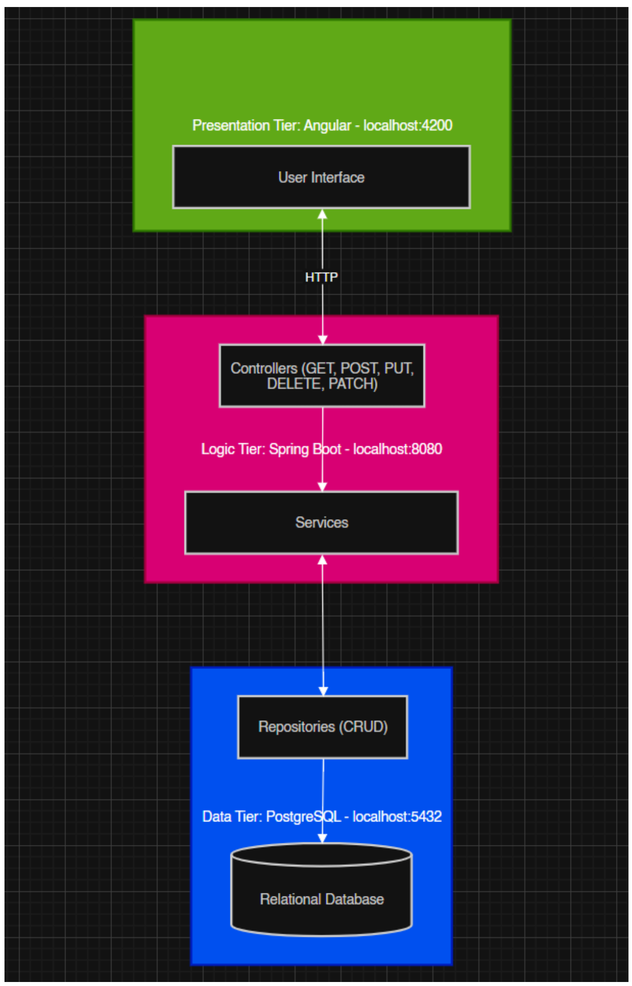
3. System Architectural Design

3.1 Architectural Pattern Description

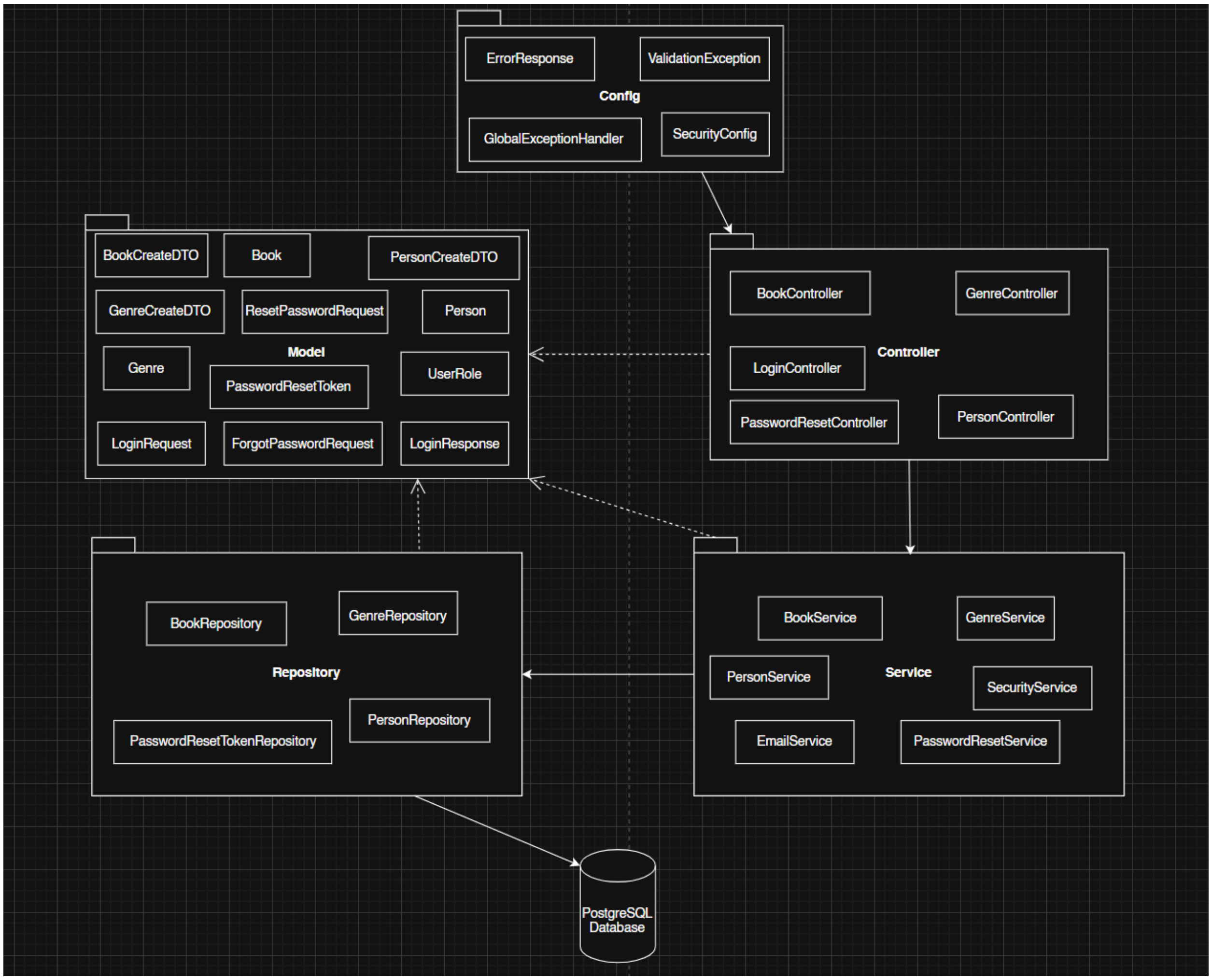
The application is designed in a Layered Architecture divided into 3 tiers:

- **Presentation Tier (Client):** Implemented using Angular 19. This tier handles the user interface and user interactions, communicating with the backend via RESTful APIs.
- **Logic Tier (Server):** Implemented using Java with Spring Boot. This serves as the middle layer that processes business logic, validates data, and coordinates between the UI and the database.
- **Data Tier (Database):** Implemented using PostgreSQL. This tier is responsible for persistent storage of persons, books, and genres.

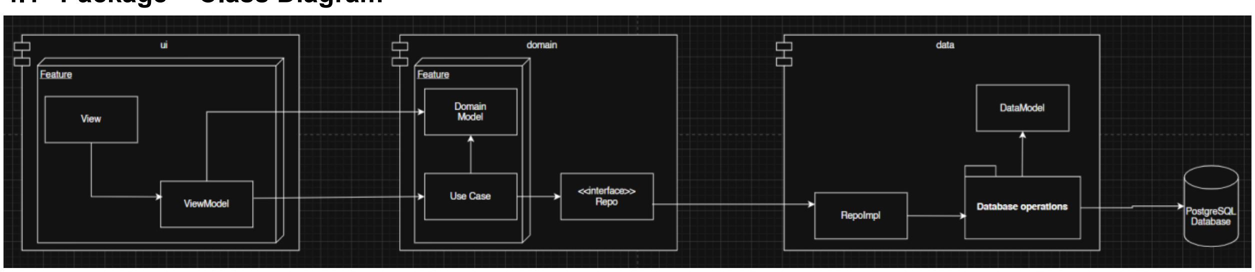
3.2 Diagrams



4. Class Design

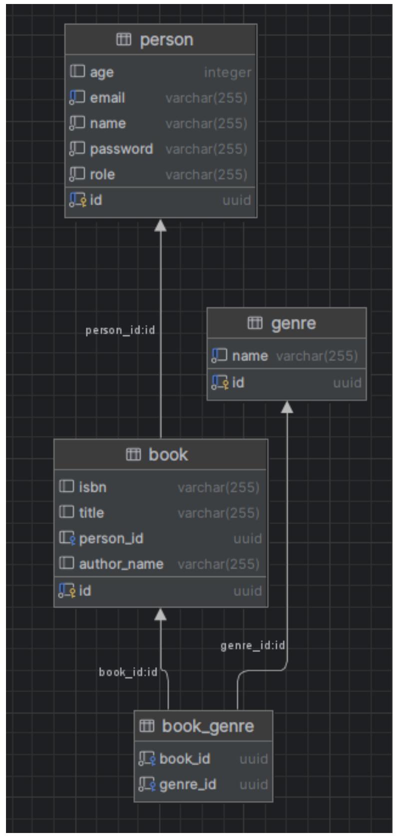


4.1 Package + Class Diagram



5. Data Model

The data model for the library management application consists of five primary entities structured to support user ,book and genre management. The relational schema utilizes primary keys (PK) for entity identification and foreign keys (FK) to establish One-to-Many (1:n) and Many-to-Many (n:m) relationships.



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name varchar(255)
password varchar(255)
role varchar(255)
id uuid
person_id:id
genre
name varchar(255)
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book
isbn varchar(255)
title varchar(255)
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author_name varchar(255)
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book_id uuid
genre_id uuid